

Spring REST - Employee and Department Web Service

Create static employee list data using spring xml configuration

employee.xml (src/main/resources)

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="dept_it" class="com.cognizant.springlearn.Department">
        <property name="id" value="1" />
        <property name="name" value="IT" />
    </bean>

    <bean id="dept_hr" class="com.cognizant.springlearn.Department">
        <property name="id" value="2" />
        <property name="name" value="HR" />
    </bean>

    <bean id="skill_java" class="com.cognizant.springlearn.Skill">
        <property name="id" value="1" />
        <property name="name" value="Java" />
    </bean>

    <bean id="skill_angular" class="com.cognizant.springlearn.Skill">
        <property name="id" value="2" />
        <property name="name" value="Angular" />
    </bean>

    <bean id="skill_sql" class="com.cognizant.springlearn.Skill">
        <property name="id" value="3" />
        <property name="name" value="SQL" />
    </bean>

    <bean id="emp1" class="com.cognizant.springlearn.Employee">
        <property name="id" value="1" />
        <property name="name" value="Arun Kumar" />
        <property name="email" value="arun.kumar@company.com" />
        <property name="department" ref="dept_it" />
        <property name="skilllist">
            <list>
                <ref bean="skill_java" />
                <ref bean="skill_sql" />
            </list>
        </property>
    </bean>

    <bean id="emp2" class="com.cognizant.springlearn.Employee">
        <property name="id" value="2" />
        <property name="name" value="Divya Menon" />
        <property name="email" value="divya.menon@company.com" />
        <property name="department" ref="dept_it" />
        <property name="skilllist">
            <list>
                <ref bean="skill_angular" />
            </list>
        </property>
    </bean>

</beans>
```

```

        </property>
    </bean>

    <bean id="emp3" class="com.cognizant.springlearn.Employee">
        <property name="id" value="3" />
        <property name="name" value="Karthik Raja" />
        <property name="email" value="karthik.raja@company.com" />
        <property name="department" ref="dept_hr" />
        <property name="skillList">
            <list>
                <ref bean="skill_sql" />
            </list>
        </property>
    </bean>

    <bean id="emp4" class="com.cognizant.springlearn.Employee">
        <property name="id" value="4" />
        <property name="name" value="Priya Sundaram" />
        <property name="email" value="priya.sundaram@company.com" />
        <property name="department" ref="dept_hr" />
        <property name="skillList">
            <list>
                <ref bean="skill_java" />
                <ref bean="skill_angular" />
            </list>
        </property>
    </bean>

    <bean id="employeeList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <ref bean="emp1" />
                <ref bean="emp2" />
                <ref bean="emp3" />
                <ref bean="emp4" />
            </list>
        </constructor-arg>
    </bean>

    <bean id="departmentList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <ref bean="dept_it" />
                <ref bean="dept_hr" />
            </list>
        </constructor-arg>
    </bean>

</beans>

```

Department.java

```

package com.cognizant.springlearn;

public class Department {

    private int id;
    private String name;

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
}

```

```

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    @Override
    public String toString() {
        return "Department [id=" + id + ", name=" + name + "]";
    }
}

```

Skill.java

```

package com.cognizant.springlearn;

public class Skill {

    private int id;
    private String name;

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    @Override
    public String toString() {
        return "Skill [id=" + id + ", name=" + name + "]";
    }
}

```

Employee.java

```

package com.cognizant.springlearn;

import java.util.List;

public class Employee {

    private int id;
    private String name;
    private String email;
    private Department department;
    private List<Skill> skillList;

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    public Department getDepartment() { return department; }
    public void setDepartment(Department department) { this.department = department; }

    public List<Skill> getSkillList() { return skillList; }
    public void setSkillList(List<Skill> skillList) { this.skillList = skillList; }

    @Override

```

```

    public String toString() {
        return "Employee [id=" + id + ", name=" + name + ", email=" + email + "]";
    }
}

```

EmployeeDao.java

```

package com.cognizant.springlearn.dao;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Repository;
import com.cognizant.springlearn.Employee;

@Repository
public class EmployeeDao {

    private static final Logger LOGGER = LoggerFactory.getLogger(EmployeeDao.class);

    private static List<Employee> EMPLOYEE_LIST;

    @SuppressWarnings("unchecked")
    public EmployeeDao() {
        LOGGER.debug("Inside EmployeeDao Constructor");
        ApplicationContext context = new ClassPathXmlApplicationContext("employee.xml");
        EMPLOYEE_LIST = (List<Employee>) context.getBean("employeeList");
    }

    public List<Employee> getAllEmployees() {
        LOGGER.info("START");
        LOGGER.info("END");
        return EMPLOYEE_LIST;
    }
}

```

Create REST service to gets all employees

EmployeeService.java

```

package com.cognizant.springlearn.service;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import com.cognizant.springlearn.Employee;
import com.cognizant.springlearn.dao.EmployeeDao;

@Service
public class EmployeeService {

    private static final Logger LOGGER = LoggerFactory.getLogger(EmployeeService.class);

    @Autowired
    private EmployeeDao employeeDao;
}

```

```

    @Transactional
    public List<Employee> getAllEmployees() {
        LOGGER.info("START");
        List<Employee> employeeList = employeeDao.getAllEmployees();
        LOGGER.info("END");
        return employeeList;
    }
}

```

EmployeeController.java

```

package com.cognizant.springlearn.controller;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Employee;
import com.cognizant.springlearn.service.EmployeeService;

@RestController
public class EmployeeController {

    private static final Logger LOGGER = LoggerFactory.getLogger(EmployeeController.class);

    @Autowired
    private EmployeeService employeeService;

    @GetMapping("/employees")
    public List<Employee> getAllEmployees() {
        LOGGER.info("START");
        List<Employee> employeeList = employeeService.getAllEmployees();
        LOGGER.info("END");
        return employeeList;
    }
}

```

Output - Postman GET http://localhost:8083/employees

```

Status: 200 OK
Body:
[
  {
    "id": 1,
    "name": "Arun Kumar",
    "email": "arun.kumar@company.com",
    "department": { "id": 1, "name": "IT" },
    "skillList": [ { "id": 1, "name": "Java" }, { "id": 3, "name": "SQL" } ]
  },
  {
    "id": 2,
    "name": "Divya Menon",
    "email": "divya.menon@company.com",
    "department": { "id": 1, "name": "IT" },
    "skillList": [ { "id": 2, "name": "Angular" } ]
  },
  {
    "id": 3,
    "name": "Karthik Raja",
    "email": "karthik.raja@company.com",

```

```

    "department": { "id": 2, "name": "HR" },
    "skillList": [ { "id": 3, "name": "SQL" } ]
  },
  {
    "id": 4,
    "name": "Priya Sundaram",
    "email": "priya.sundaram@company.com",
    "department": { "id": 2, "name": "HR" },
    "skillList": [ { "id": 1, "name": "Java" }, { "id": 2, "name": "Angular" } ]
  }
]

```

Output - console log

```

240715|13:02:11.204|main          |DEBUG|EmployeeDao          |
<init>|Inside EmployeeDao Constructor
240715|13:02:41.116|http-nio-8083-exec-1|INFO  |EmployeeController    |
getAllEmployees|START
240715|13:02:41.118|http-nio-8083-exec-1|INFO  |EmployeeService       |
getAllEmployees|START
240715|13:02:41.119|http-nio-8083-exec-1|INFO  |EmployeeDao           |
getAllEmployees|START
240715|13:02:41.119|http-nio-8083-exec-1|INFO  |EmployeeDao           |
getAllEmployees|END
240715|13:02:41.120|http-nio-8083-exec-1|INFO  |EmployeeService       |
getAllEmployees|END
240715|13:02:41.121|http-nio-8083-exec-1|INFO  |EmployeeController    |
getAllEmployees|END

```

Create REST service for department

DepartmentDao.java

```

package com.cognizant.springlearn.dao;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Repository;
import com.cognizant.springlearn.Department;

@Repository
public class DepartmentDao {

    private static final Logger LOGGER = LoggerFactory.getLogger(DepartmentDao.class);

    private static List<Department> DEPARTMENT_LIST;

    @SuppressWarnings("unchecked")
    public DepartmentDao() {
        LOGGER.debug("Inside DepartmentDao Constructor");
        ApplicationContext context = new ClassPathXmlApplicationContext("employee.xml");
        DEPARTMENT_LIST = (List<Department>) context.getBean("departmentList");
    }

    public List<Department> getAllDepartments() {
        LOGGER.info("START");
        LOGGER.info("END");
        return DEPARTMENT_LIST;
    }
}

```

```
}  
}
```

DepartmentService.java

```
package com.cognizant.springlearn.service;  
  
import java.util.List;  
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.stereotype.Service;  
import org.springframework.transaction.annotation.Transactional;  
import com.cognizant.springlearn.Department;  
import com.cognizant.springlearn.dao.DepartmentDao;  
  
@Service  
public class DepartmentService {  
  
    private static final Logger LOGGER = LoggerFactory.getLogger(DepartmentService.class);  
  
    @Autowired  
    private DepartmentDao departmentDao;  
  
    @Transactional  
    public List<Department> getAllDepartments() {  
        LOGGER.info("START");  
        List<Department> departmentList = departmentDao.getAllDepartments();  
        LOGGER.info("END");  
        return departmentList;  
    }  
}
```

DepartmentController.java

```
package com.cognizant.springlearn.controller;  
  
import java.util.List;  
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.web.bind.annotation.GetMapping;  
import org.springframework.web.bind.annotation.RestController;  
import com.cognizant.springlearn.Department;  
import com.cognizant.springlearn.service.DepartmentService;  
  
@RestController  
public class DepartmentController {  
  
    private static final Logger LOGGER =  
        LoggerFactory.getLogger(DepartmentController.class);  
  
    @Autowired  
    private DepartmentService departmentService;  
  
    @GetMapping("/departments")  
    public List<Department> getAllDepartments() {  
        LOGGER.info("START");  
        List<Department> departmentList = departmentService.getAllDepartments();  
        LOGGER.info("END");  
        return departmentList;  
    }  
}
```

```
}
```

Output - Postman GET http://localhost:8083/departments

Status: 200 OK

Body:

```
[  
  { "id": 1, "name": "IT" },  
  { "id": 2, "name": "HR" }  
]
```

Output - console log

```
240715|13:09:57.402|http-nio-8083-exec-2|INFO |DepartmentController |  
getAllDepartments|START  
240715|13:09:57.403|http-nio-8083-exec-2|INFO |DepartmentService |  
getAllDepartments|START  
240715|13:09:57.403|http-nio-8083-exec-2|INFO |DepartmentDao |  
getAllDepartments|START  
240715|13:09:57.404|http-nio-8083-exec-2|INFO |DepartmentDao |  
getAllDepartments|END  
240715|13:09:57.404|http-nio-8083-exec-2|INFO |DepartmentService |  
getAllDepartments|END  
240715|13:09:57.405|http-nio-8083-exec-2|INFO |DepartmentController |  
getAllDepartments|END
```